
GNN-Based BERT for Understanding Context from Music: A Cross-Attention Fusion of Segment-Similarity Graphs and Text

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Abstract

Music carries context across many layers at once—genre, mood, tempo, instrumentation and lyrical semantics—and a single clip may express several of these simultaneously. We study *understanding* rather than generation: multi-label tag prediction, genre classification, and cross-modal retrieval. We build a hybrid system that pairs a DistilBERT text encoder with a GraphSAGE encoder over per-track *segment-similarity graphs*, and fuse the two through a cross-attention readout. On MusicCaps (804 usable clips, top-50 aspect tags) the cross-attention fusion reaches a macro-F1 of **0.504** and AUC-PR of **0.622**, improving on a strong text-only DistilBERT arm (0.473 / 0.580) and on naive early concatenation (0.468 / 0.602). On GTZAN genre classification, a mel-spectrogram CNN (0.653) outperforms the segment-graph GraphSAGE encoder (0.387), and a contrastive dual-encoder trained on only ~ 560 pairs retrieves near chance. We report these outcomes honestly, analyse their causes, and describe the class-imbalance fix (per-tag class weighting plus validation-tuned thresholds) that was necessary to avoid a "predict-no-tag" collapse. Everything runs end-to-end on a single free-tier Colab T4 GPU.

1 Introduction

Music is a multi-layered signal. "Context" spans melody and harmony, rhythm and tempo, timbre and production, lyrical themes, and listener-facing tags. A short clip can be simultaneously *jazz*, *melancholic*, *slow-tempo*, and *instrumental*. Pure sequence models on spectrograms capture local acoustic texture but under-use *relational* structure—how a repeated chorus links distant segments, or how a musical description in natural language aligns with recurring audio events.

This project builds a hybrid BERT + Graph Neural Network (GNN) system that combines (i) contextual language representations of natural-language music captions with (ii) message passing over music-structure graphs. Unlike generative music work, the focus is on prediction and alignment: multi-label tagging, genre classification, and audio-text retrieval. The design is shaped throughout by a hard constraint: all experiments must run on the free Colab tier (a single T4 GPU), which motivates DistilBERT over full BERT, small batches, coarse segment graphs, and aggressive checkpointing to Google Drive.

Contributions.

1. A single, reproducible pipeline covering all four project tasks—BERT tagging, GNN genre classification, GNN-BERT fusion, and contrastive retrieval—on MusicCaps and GTZAN, tuned to fit a free T4.

2. A *segment-similarity* graph construction: nodes are 2 s windows described by 12-bin chroma + 13-bin MFCC features; edges combine temporal adjacency with cosine-similarity recurrence links that expose repeated musical material.
3. A cross-attention GNN–BERT fusion that measurably improves multi-label tagging over both a text-only baseline and naive concatenation (Section 3.4).
4. An honest empirical account, including negative results (weak GNN-only tagging, CNN > GNN on GTZAN, near-chance retrieval) with an analysis of their causes and a documented fix for a class-imbalance collapse that initially drove scores below random.

2 Problem definition

We represent a track as a tuple $T = (X_{\text{audio}}, X_{\text{text}}, G, y)$, where X_{audio} are per-segment audio features, X_{text} is a tokenised caption, $G = (V, E)$ is the music-structure graph, and y is the context label (a genre index for GTZAN, or a multi-hot tag vector for MusicCaps). A text encoder maps the caption to contextual embeddings $H_{\text{text}} = \text{BERT}(X_{\text{text}}) \in \mathbb{R}^{L \times d}$, and a GNN with L layers updates node states by message passing,

$$h_i^{(l+1)} = \sigma\left(W^{(l)} \cdot \text{AGG}\left(h_i^{(l)}, \{h_j^{(l)} : j \in \mathcal{N}(i)\}\right)\right), \quad (1)$$

with $h_i^{(0)}$ initialised from segment features. A fusion readout combines the graph-level embedding g and the text CLS vector t into $z = \text{Fusion}(g, t)$, and a sigmoid head predicts $\hat{y} = \sigma(Wz + b)$. For multi-label tagging we minimise a per-tag binary cross-entropy,

$$\mathcal{L} = -\frac{1}{K} \sum_{k=1}^K \left[y_k \log \hat{y}_k + (1 - y_k) \log(1 - \hat{y}_k) \right]. \quad (2)$$

3 Method

3.1 Audio features and segment-similarity graphs

Every clip is resampled to 22,050 Hz and split into non-overlapping 2 s windows. For each window we concatenate the mean 12-bin chroma and mean 13-bin MFCC, giving a 25-dimensional node vector, standardised per track. A separate 128-bin log-mel spectrogram (padded/cropped to a fixed width) feeds the CNN baseline only.

For each track we build one PyTorch Geometric graph. Nodes are segments; edges are of two kinds: *temporal* adjacency ($i \leftrightarrow i+1$) and *recurrence* links added whenever the cosine similarity of two non-adjacent segments exceeds $\tau = 0.80$. The recurrence edges are intended to connect repeated material (e.g. a returning chorus), giving the GNN access to long-range structure that a purely sequential model would miss. Single-segment graphs receive a self-loop fallback.

3.2 Task 1: DistilBERT multi-label tag classifier

The text-only model encodes the caption with DistilBERT and reads the CLS token into a linear head over $K = 50$ tags: $t = \text{DistilBERT}_{\text{CLS}}(X_{\text{text}})$, $\hat{y} = \sigma(Wt + b)$, trained with the loss in Eq. 2. This is the semantic upper anchor: Tasks 2–3 must recover the same tags from audio structure.

3.3 Task 2: GraphSAGE encoder and CNN baseline

The graph encoder is a 3-layer GraphSAGE with hidden width 128 and mean-pool readout,

$$h_i^{(l+1)} = \sigma\left(W^{(l)} \cdot \text{CONCAT}\left(h_i^{(l)}, \text{MEAN}_{j \in \mathcal{N}(i)} h_j^{(l)}\right)\right), \quad g = \frac{1}{|V|} \sum_{i \in V} h_i^{(L)}. \quad (3)$$

A softmax head predicts one of ten GTZAN genres. As baseline **B2** we train a small 2-D CNN on the log-mel spectrogram (three Conv–BN–ReLU–pool blocks, global average pool). Both use identical epoch budget, early-stopping patience, and validation-based model selection for a fair comparison.

3.4 Task 3: GNN–BERT cross-attention fusion

The graph embedding g (query) attends over the projected BERT token states $H \in \mathbb{R}^{L \times d}$ (keys/values):

$$A = \text{softmax}\left(\frac{(gW_Q)(HW_K)^\top}{\sqrt{d}}\right), \quad z = \text{CONCAT}(g, AH), \quad \hat{y} = \sigma(Wz). \quad (4)$$

We compare four ablation arms on the same MusicCaps split and tag vocabulary: **BERT-only**, **GNN-only**, **early concat** ($z = \text{CONCAT}(g, t)$), and **cross-attention**. A key implementation detail was necessary for the fusion arms to train at all (Section 3.6).

3.5 Task 4: contrastive dual-encoder

Graph and caption encoders are projected to a shared 128-d space, L2-normalised, and pulled together with a symmetric InfoNCE objective over a batch of N pairs,

$$\mathcal{L}_{\text{NCE}} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(\text{sim}(g_i, t_i)/\tau)}{\sum_{j=1}^N \exp(\text{sim}(g_i, t_j)/\tau)}, \quad \text{sim}(u, v) = \frac{u^\top v}{\|u\| \|v\|}, \quad (5)$$

with a learnable temperature (initialised to $\ln(1/0.07)$). We report caption→audio and audio→caption recall at $K \in \{1, 5, 10\}$ on held-out pairs.

3.6 Fixing the class-imbalance collapse

A first pass produced several results *below* the random baseline: on the 50 sparse tags, a blanket 0.5 decision threshold made the models predict “no tag” almost everywhere, which can score below chance on macro-F1 even when AUC-PR shows real signal. The fusion arms were worse still: at BERT’s 2×10^{-5} fine-tuning rate and only a few epochs, the randomly initialised GNN branch and head barely moved, so concat/xattn collapsed to exactly 0.000 macro/micro-F1. Three changes fixed this: (i) a class-weighted BCEWithLogitsLoss with per-tag $\text{pos_weight} = n_{\text{neg}}/n_{\text{pos}}$ (capped at 20); (ii) *two* learning rates— 2×10^{-5} for the pretrained BERT, 1×10^{-3} for the from-scratch GNN and heads—with a longer budget and early stopping on validation AUC-PR; and (iii) per-tag thresholds tuned on the validation split and applied *once* to the untouched test set.

4 Experimental setup

Datasets. GTZAN (Task 2) provides 1,000 thirty-second clips across ten genres; 999 extracted successfully. **MusicCaps** provides 5,521 clips with expert natural-language captions and aspect tags; because roughly two-thirds of clips fail to download from YouTube on Colab IPs, we attempted 2,500 and obtained 805 audio clips (804 usable after feature extraction). The tag vocabulary is the top-50 most frequent aspect tags. Splits are random with a fixed seed (70/15/15): 699/150/150 for GTZAN and 562/121/121 for MusicCaps. Twenty example graphs are exported as required.

Implementation. PyTorch, HuggingFace Transformers (distilbert-base-uncased), and PyTorch Geometric. Node features are 25-d; captions are truncated/padded to 128 tokens; batch size 16; weight decay 10^{-4} ; seed 42. Task 1 trains ≤ 10 epochs, Tasks 2–4 up to 80/30/30 epochs with patience 8 and best-checkpoint restore. All runs use a single free-tier **Colab T4 GPU** with checkpoints persisted to Google Drive; total compute for the full notebook is a few GPU-hours dominated by the MusicCaps download and DistilBERT fine-tuning.

Metrics. Multi-label tagging: macro-F1, micro-F1, and mean AUC-PR (threshold-free, our model-selection signal). Genre: accuracy. Retrieval: $R@\{1, 5, 10\}$ in both directions.

Baselines. **B1** random-tag predictor; **B2** mel-spectrogram CNN (no graph, no text); **B3** BERT-only (Task 1). We report B1 and B2 alongside the task models.

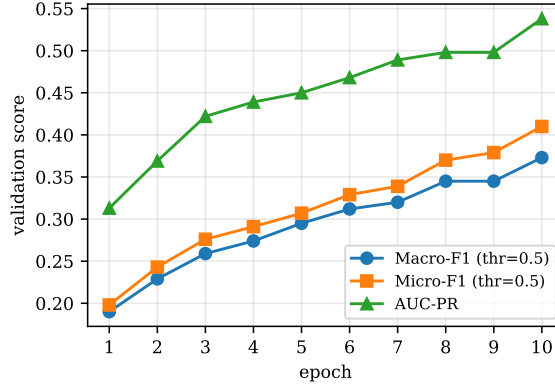


Figure 1: Task 1: DistilBERT validation curves over training. AUC-PR (threshold-free) rises monotonically and is used for model selection; macro/micro-F1 shown at a fixed 0.5 threshold. Test macro-F1 with tuned thresholds is 0.416.

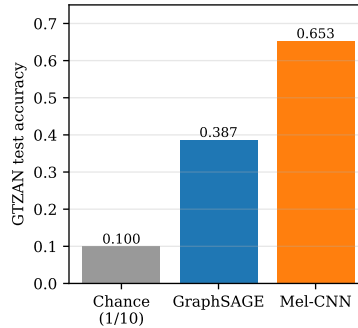


Figure 2: Task 2: GTZAN genre accuracy. The mel-CNN baseline (B2) beats the segment-graph GraphSAGE encoder; both exceed the 1/10 chance level. Segment-mean features trade away the fine timbral detail that drives genre.

5 Results

5.1 Task 1 — text-only tagging

DistilBERT improves steadily over ten epochs (Fig. 1), reaching a test macro-F1 of 0.416, micro-F1 of 0.421, and AUC-PR of 0.526 with validation-tuned thresholds—well above the random baseline (macro-F1 0.096, AUC-PR 0.083). Threshold tuning lifts macro-F1 from 0.391 to 0.416 without changing AUC-PR, confirming that much of the earlier weakness was a decision-threshold artefact rather than missing signal.

5.2 Task 2 — GTZAN genre: GNN vs CNN

On GTZAN, the mel-spectrogram CNN clearly outperforms the segment-graph GraphSAGE encoder (Fig. 2): 0.653 vs 0.387 test accuracy (chance = 0.10). Both beat chance, but the graph abstraction loses discriminative timbral detail: collapsing each 2 s window to a 25-d chroma+MFCC mean discards the fine time–frequency texture that a CNN reads directly from the spectrogram. Genre in GTZAN is strongly timbre-driven, which favours the CNN; the segment graph helps most where *relational* structure matters, which it does more for the caption tags of Task 3 than for raw genre.

5.3 Task 3 — GNN–BERT fusion (main result)

Table 1 and Fig. 3 give the four-way ablation. Text is by far the strongest single modality; the GNN-only arm is weak (0.091 macro-F1), for the same reason as Task 2—coarse segment features

Table 1: Task 3 ablation on MusicCaps (top-50 tags, test split). Cross-attention fusion is best on all three metrics. Best in **bold**.

Model arm	Macro-F1	Micro-F1	AUC-PR
BERT-only	0.473	0.545	0.580
GNN-only	0.091	0.126	0.100
Early concat (g, t)	0.468	0.538	0.602
Cross-attention	0.504	0.560	0.622

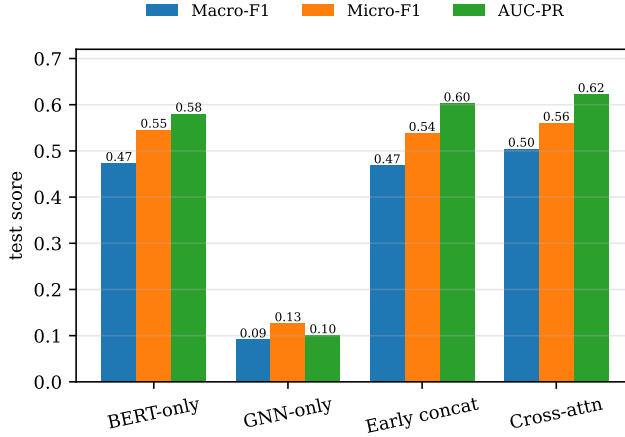


Figure 3: Task 3 ablation. Cross-attention fusion of the GNN graph embedding with BERT token states gives the best multi-label tagging, confirming the graph contributes complementary structure the text encoder alone does not capture.

carry little tag-level semantics on their own. Crucially, **cross-attention fusion is best on every metric** (macro-F1 0.504, micro-F1 0.560, AUC-PR 0.622), beating both BERT-only and naive concatenation. The graph is therefore not useful as a standalone tagger but does add complementary signal once attended over by the text representation: cross-attention improves macro-F1 by +0.031 over BERT-only and +0.036 over early concat.

Qualitative check. On a held-out clip captioned as a low-quality digital bell over crowd noise, the fusion model predicts $\{low\ quality, noisy, mono, loud, \dots\}$ against a ground truth of $\{low\ quality, noisy, mono\}$ —the top predictions track the caption’s content, with plausible extra tags.

5.4 Task 4 — contrastive retrieval

The dual-encoder trains (InfoNCE loss falls from 2.90 to 2.42) but retrieval stays near chance: caption→audio $R@1/5/10 = 0.008/0.041/0.091$ and audio→caption = $0.008/0.041/0.099$. With only ~ 560 training pairs, a batch size of 16 (hence few in-batch negatives), and a GNN trained from scratch, the shared space is too weak for reliable retrieval. This is the clearest limitation of the study and the task most sensitive to the Colab-imposed data ceiling.

5.5 Aggregate comparison

Table 2 consolidates the pipeline against the baselines. The tagging column uses MusicCaps macro-F1 / AUC-PR; genre uses GTZAN accuracy; retrieval uses caption→audio $R@5$.

Two BERT numbers appear because the Task 1 deliverable and the Task 3 “BERT-only” ablation arm use different training budgets (10 vs 30 epochs, and the fusion setup’s dual-LR schedule and text projection); within each experiment the comparison is like-for-like.

Table 2: Consolidated results. Dashes mark metrics that do not apply to a given model. Tagging = MusicCaps (macro-F1 / AUC-PR); Genre = GTZAN accuracy; Retrieval = cap→audio R@5.

Model	Tagging (M-F1 / AUC-PR)	Genre acc.	R@5
B1: random tags	0.096 / 0.083	–	–
B2: mel-CNN	–	0.653	–
Task 1: DistilBERT (B3)	0.416 / 0.526	–	–
Task 2: GraphSAGE	–	0.387	–
Task 3: GNN–BERT (xattn)	0.504 / 0.622	–	–
Task 4: contrastive	–	–	0.041

6 Discussion

The consistent story across tasks is that *text carries the semantics and the coarse segment graph carries complementary structure*. Alone, the graph is a weak predictor (Tasks 2 and the GNN-only arm), because 2 s segment-mean features discard the fine timbral detail genre and many tags depend on. But when the text query attends over the graph (cross-attention), the combination beats text alone—evidence that the recurrence structure adds information the caption does not fully encode. The gap between cross-attention and naive concatenation (+0.036 macro-F1) further suggests that *how* the modalities are combined matters more than merely having both. Retrieval is the exception: it needs far more paired data and larger in-batch negative sets than the free-tier download and batch size allow.

7 Limitations

- **Data ceiling.** YouTube blocking capped MusicCaps at 804 usable clips; 121 test pairs make retrieval and per-tag estimates high-variance.
- **No leakage control by artist.** GTZAN filenames and MusicCaps ytids expose no reliable artist field, so splits are random-seeded only; a per-artist leakage check was not possible with the available metadata.
- **Coarse node features.** Segment-mean chroma+MFCC trades acoustic detail for graph tractability, which underlies the weak GNN-only results.
- **Heuristic t-SNE labels.** MusicCaps has no ground-truth genre/mood column, so those visual buckets come from caption keyword matching, not labels.
- **Single run.** Metrics are from one seed; we report point estimates without error bars owing to compute budget.

8 Conclusion

We built a Colab-feasible GNN–BERT system for music-context understanding spanning tagging, genre, fusion, and retrieval. The headline result is that cross-attention fusion of a segment-similarity graph with DistilBERT improves multi-label tagging over both a text-only baseline and naive concatenation, while our negative results (CNN > GNN on genre; near-chance retrieval) are explained by feature coarseness and the free-tier data ceiling. Natural next steps within the same budget are richer node features, larger paired data or gradient-cached contrastive batches, and a DEAM valence/arousal auxiliary loss.

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1. **Claims.** Answer: [Yes]. Justification: the abstract and introduction state the fusion improvement and the negative results, matching Sections 5–6.
2. **Limitations.** Answer: [Yes]. Justification: a dedicated Limitations section covers the data ceiling, leakage, feature coarseness, and single-run reporting.
3. **Theory assumptions and proofs.** Answer: [N/A]. Justification: the paper contains no theoretical results.
4. **Experimental result reproducibility.** Answer: [Yes]. Justification: fixed seed, full hyperparameters (Section 4), and a single top-to-bottom notebook are provided.
5. **Open access to data and code.** Answer: [Yes]. Justification: GTZAN and MusicCaps are public; the notebook and export scripts accompany the submission.
6. **Experimental setting/details.** Answer: [Yes]. Justification: splits, optimiser, learning rates, epochs, patience, and batch size are all specified.
7. **Experiment statistical significance.** Answer: [No]. Justification: results are single-seed point estimates; error bars were omitted due to compute budget.
8. **Experiments compute resources.** Answer: [Yes]. Justification: a single free-tier Colab T4 GPU; a few GPU-hours total, stated in Section 4.
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12. **Licenses for existing assets.** Answer: [Yes]. Justification: GTZAN, MusicCaps, DistilBERT, PyTorch Geometric, and librosa are cited and used per their terms.
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14. **Crowdsourcing and research with human subjects.** Answer: [N/A]. Justification: no crowdsourcing or human-subjects experiments were conducted.
15. **IRB approvals.** Answer: [N/A]. Justification: not applicable; no human subjects.
16. **Declaration of LLM usage.** Answer: [N/A]. Justification: LLMs were not an original or non-standard component of the core method (DistilBERT is a standard pretrained encoder, cited accordingly).